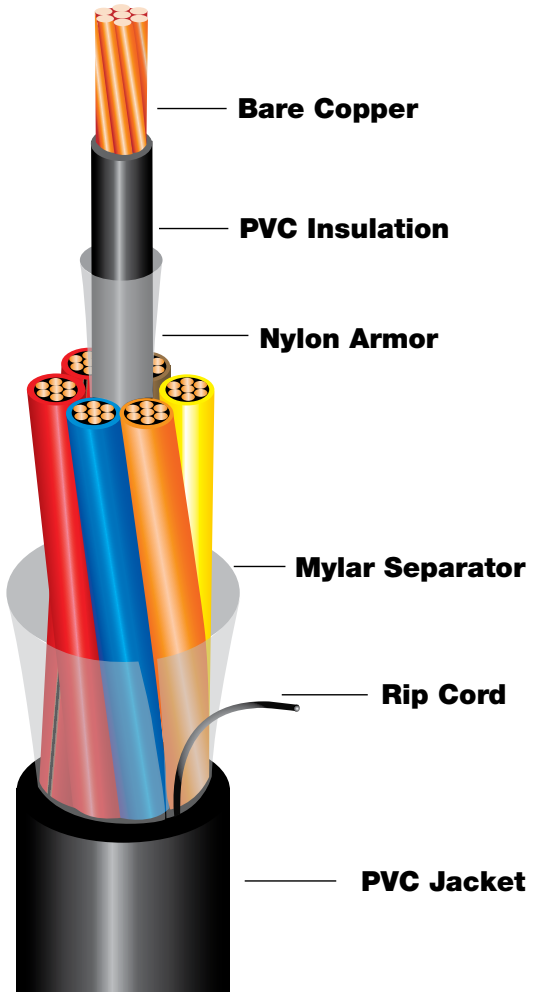
TYPE TC-ER - INDUSTRIAL CONTROL CABLE

PVC/Nylon Insulation with Overall PVC Jacket
18 - 10 AWG • 600 Volts • 90°C Dry/Wet



Cable Identification

18-16 AWG
"ADVANCED DIGITAL CABLE, INC. XX AWG XX TYPE TFN CDRS (UL) TYPE TC OR TC-ER 90C SUN RES DIR BUR 600V FT4/ IEEE1202 E195597 MADE IN USA"

14-10 AWG
"ADVANCED DIGITAL CABLE, INC. XX AWG XX TYPE THHN CDRS (UL) TYPE TC OR TC-ER 90C SUN RES DIR BUR 600V FT4/IEEE1202 E195597 MADE IN USA"

MADE IN THE USA

Description

ADC's Type TC-ER multi-conductor cables have a TFN or THHN/THWN (PVC/Nylon) insulation with an overall gas and oil resistant Polyvinyl Chloride (PVC) jacket.

Applications

Suitable for use in Class 1 or 2, Division 2 hazardous locations and for installation in trays, wireways, troughs, channels, ducts and conduit. Expressly approved for direct burial, wet or dry locations and outdoors in cable trays where sunlight resistant rating is required. Intended for control, power, lighting, telemetering, signals and relay or traffic control.

Construction

Conductors: Soft Drawn Annealed Bare Copper per ASTM B-3 Concentric 7 strand. Concentric 19 strand available upon request.

Insulation: Thickness: UL-83 table 10 for THHN/THWN, UL-66 table 4.7 for TFN.

Conductor Jacket: Thickness: UL-83 table 13 for THHN/THWN, UL-66 paragraph 9.1 for TFN.

Cabling: Three or more conductors are assembled with fillers in the core as needed. Two conductors are assembled flat parallel or round with fillers as needed.

Separator: Mylar

Overall Jacket: A black, flame resistant, Polyvinyl Chloride (PVC) jacket is extruded over the assembly. The surface profile shall approximate that of the interior assembly. A nylon rip cord shall be inserted under the jacket for ease of stripping.

Color Code: ICEA Method 1, Table E-2 Standard. ICEA Method 1 Tables E-1, E-3 & Method 4.

Industry Listings & Standards

UL Listed as TC-ER per UL Standard 1277
OSHA Acceptable
NEC Articles 392 & 336
CSA FT4
IEEE 1202 70,000 BTU Flame Test
ASTM - All Applicable Standards

UL LISTED

RoHS COMPLIANT

ADC ADVANCED DIGITAL CABLE INC.
Phone: (800) 343 2579 • fax: (828) 389 3922 • www.adcable.com

Rev 030613

PVC/Nylon Insulation with Overall PVC Jacket
18 - 10 AWG • 600 Volts • 90°C Dry/Wet

Conductor Data

Size AWG	Strands	PVC Insulation Thickness (Mils)	Nylon Armor (Mils)	Approximate O.D. (Inches)
18	7	15	5	.086
16	7	15	5	.098
14	7	15	5	.113
12	7	15	5	.132
10	7	20	5	.167

Cable Data

	18 AWG				16 AWG				14 AWG				12 AWG				10 AWG			
No. of Conds.	Part No.	OA Jkt. Thick. (MILS)	Appr. O.D. (IN)	Appr. Weight Lbs./ M Ft.	Part No.	OA Jkt. Thick. (MILS)	Appr. O.D. (IN)	Appr. Weight Lbs./ M Ft.	Part No.	OA Jkt. Thick. (MILS)	Appr. O.D. (IN)	Appr. Weight Lbs./ M Ft.	Part No.	OA Jkt. Thick. (MILS)	Appr. O.D. (IN)	Appr. Weight Lbs./ M Ft.	Part No.	OA Jkt. Thick. (MILS)	Appr. O.D. (IN)	Appr. Weight Lbs./ M Ft.
2 Flat	6802F	45	.176x .262	41	6602F	45	.188x .286	49	6402F	45	.203x .316	64	6202F	45	.222x .360	83	6102F	45	.257x .424	115
2 Round	6802	45	.266	46	6602	45	.290	54	6402	45	.320	71	6202	45	.358	92	6102	45	.428	127
3	6803	45	.279	50	6603	45	.305	66	6403	45	.337	87	6203	45	.378	113	6103	45	.453	167
4	6804	45	.301	60	6604	45	.330	79	6404	45	.366	107	6204	45	.412	145	6104	45	.496	212
5	6805	45	.322	71	6605	45	.359	94	6405	45	.399	129	6205	45	.450	175	6105	60	.575	269
6	6806	45	.348	85	6606	45	.388	109	6406	45	.433	147	6206	45	.490	199	6106	60	.625	317
7	6807	45	.348	89	6607	45	.388	118	6407	45	.433	162	6207	45	.490	223	6107	60	.625	352
8	6808	45	.375	99	6608	45	.418	133	6408	45	.468	184	6208	60	.561	268	6108	60	.697	399
9	6809	45	.405	112	6609	45	.449	147	6409	45	.503	221	6209	60	.602	304	6109	60	.727	445
10	6810	45	.435	121	6610	45	.486	162	6410	60	.576	237	6210	60	.652	327	6110	60	.792	490
12	6812	45	.451	156	6612	45	.501	202	6412	60	.593	281	6212	60	.672	388	6112	60	.817	579
15	6815	45	.498	169	6615	60	.585	243	6415	60	.655	340	6215	60	.744	466	6115	80	.949	750
19	6819	60	.554	220	6619	60	.614	296	6419	60	.689	408	6219	60	.784	581	6119	80	.999	918
25	6825	60	.640	279	6625	60	.712	379	6425	60	.802	526	6225	80	.956	796	6120	80	1.051	975

The information contained on this specification is intended to be used a guide in product selection and is believed to be reliable.
ADC has made every effort to ensure the data shown above is accurate at the time of publication. This specification is subject to change anytime without notice.